

starting with mobility and mechanical design

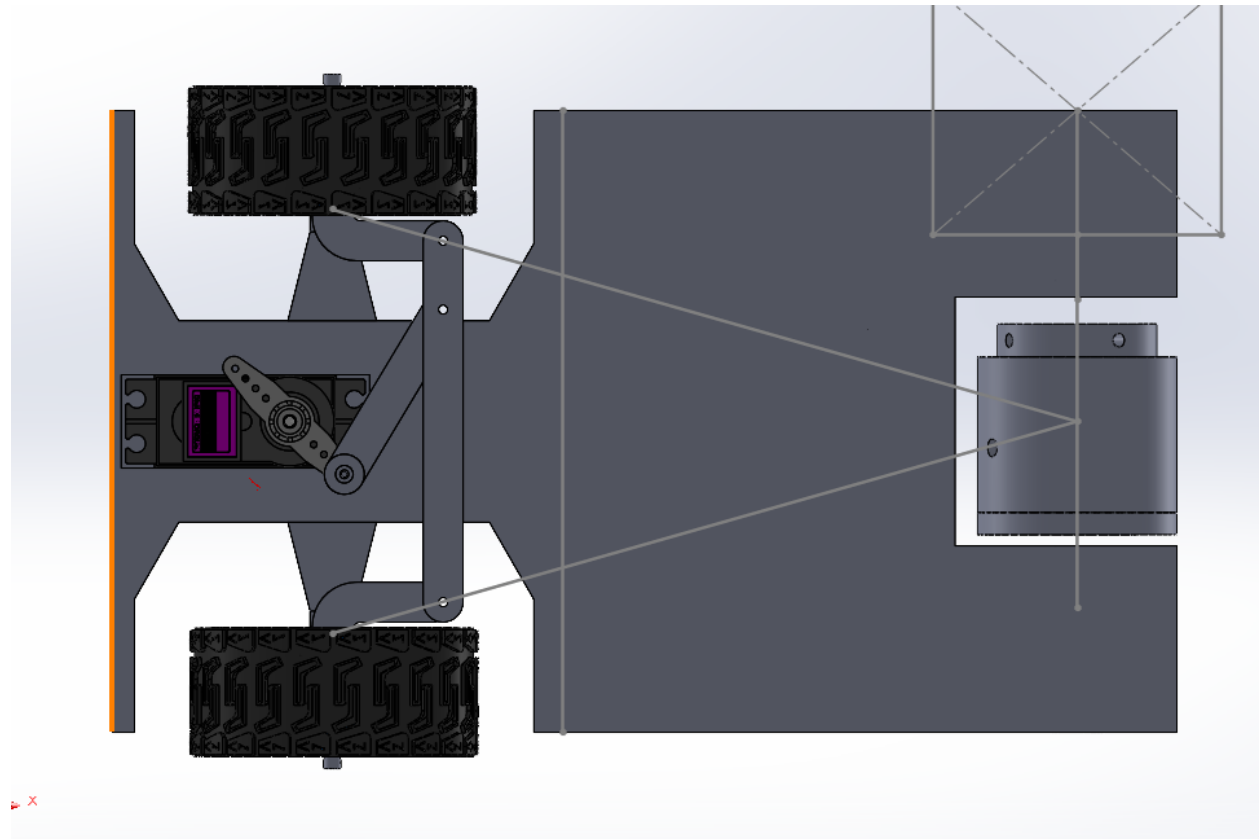
We decided we wanted to design and make our own chasis as opposed to buying one ready, it seemed far more customizable and personal, despite the attention it required, it was definitely worth it.

Then we had to choose what the footprint of our robot would look like, for starters, we already had a raspberry pi and wanted to accommodate for that, we want ease of movement and accessibility, the robot cant be too wide, or too long, so we decided on 24x12cm, plenty of space and clean movement ranges.

Our 3d printer was a Kodama trinus, it can print only pla and has a print bed of about 12x12cm which is quite small but conveniently fit half of our robots length allowing us to print the chassis in 2 halves.



next we had to design a steering system, the most reliable was an Ackermann steering system, Ackermann steering systems are extremely diverse, we chose the simplest to implement, a cross link between the Ackermann arms of both wheel knuckles, and an arm between the servo horn and the main link, allowing the servo to smoothly move the bars, connecting all of these links are simple m2 screws which provide low resistance movement.

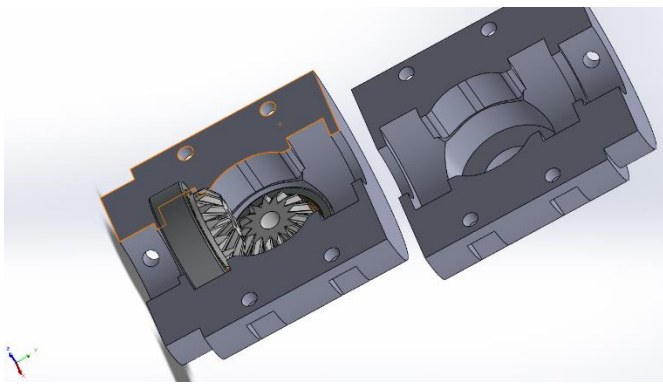


*The ends of the Ackermann arms pinpointed by the pivot of the wheel traced to the center of the back axis*

We decided the best way implement a steering and drive system was to make the back pair of wheels driven while the front only for steering, so the front two wheels had to spin freely while moving effortlessly left to right. To do this we designed a knuckle which consisted of a small axel, and a main plate, the axel has a bearing, then the wheel, then another bearing, allowing it to spin frictionlessly, the plate has 2 arms, one for the Ackermann system, and the other to connect with another screw to the chassis so it revolves around the chassis arm.

Moving the steering link was a no brainer, a servo had to be used, and with a robot of this size the only real option was the mg996r, plenty of torque, and precise control.

Moving back towards the drive, we found that the back wheels would skid during turning due to the difference in paths for both wheels, to correct for this, we designed a simple differential gear using nylon plastic gears from toys, and bearings, the differential gearbox is then driven by the motor via a belt, which is tensioned using a simple tensioner.



*Above is the design of the differential gear housing*

*To the right is the full drive train with belt tensioner.*

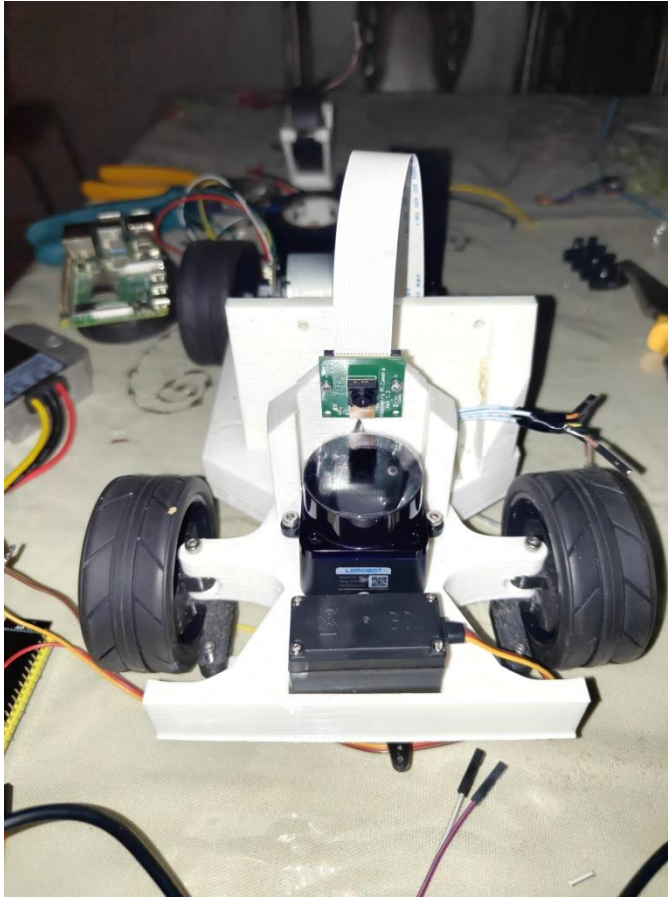


We had to choose a motor fit for driving this bulky setup, we settled on a 37mm motor we a powerful gearbox, giving us high torque and 333rpm out of the box at max speed, giving a very good range of speeds and with a high torque ceiling and tight drive train, precise movement.

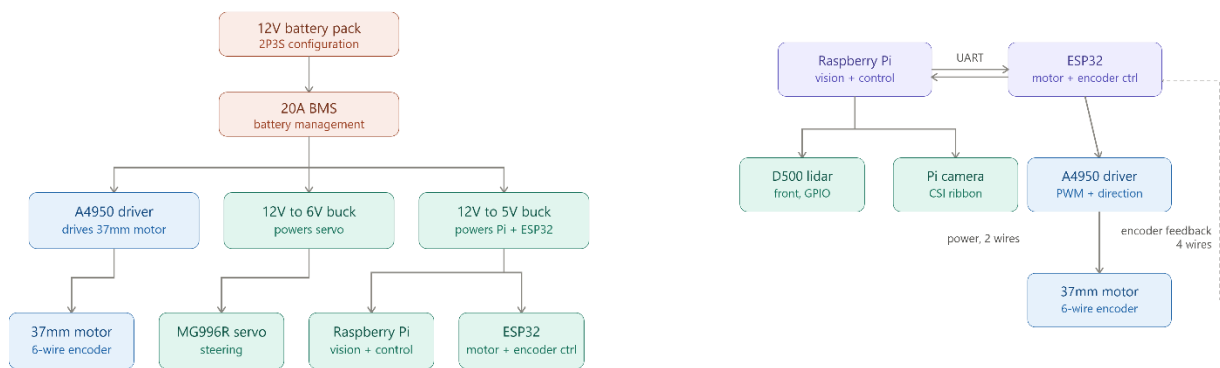
For sensors, we chose to use a powerful camera lidar setup, we chose an affordable d500 lidar we up to 50hz of turn rate giving us extremely dense point mapping and the ability to identify obstacles at their furthest. Of course a lidar alone isn't capable of identifying blocks, it requires another sensor to differentiate between the colors, we settled on using an esp camera which unfortunately arrived defected so we were forced to buy a pi camera.

Coupling both sensors gave us a system that tied the lidars ability to identify consistently the location of obstacles, and utilizing the pi's native camera system for quick analyzation we can achieve ultra fast rates of identifying obstacles.

The lidar has to be placed in a way so that the front is completely unobstructed (we aren't taking measurements from the back into consideration), the camera of course requires the same, so we the lidar was placed at the very front just behind the steering servo, and elevated on top of it on a platform is our camera angled slightly downwards to counteract any jitter or movement. Prevoisuly we had tried to use tof sensors to replicate the effectiveness of the lidar but found that tof sensors had a very hard time dealing with the black color of the arena walls whereas lidar utilized true laser technology overcoming any material and color woes.



All in all, our system had quite the high draw, with our motors in the worst cases each pulling up to 2.5 amps at stall, and our pi pulling up to 1.2, we needed a power system to keep up, which is a big negative of our bulky system, luckily our hardware was more than up to the challenge and this weight actually granted us a stable platform giving us more reliable data. We chose a 12 volt motor and a 12 volt battery to accompany, we have a 12 to 5 volt buck converter to power the pi and sensors and servo, this setup gave us plenty of headroom for any power dips from sudden power surges that can be induced by both the servo and motor, we decided to capitalize on the stability concept and implement a 2p3s system, putting us at 12 volts and 6Ah, a very large pool of energy thanks to the high density Samsung cells we were able to acquire. A 2p3s system is heavy and 3s might have been way smaller but we felt the leeway given by the higher capacity would be a good safety point, of course the high voltage paried with the capacity means it will be robust to voltage sagging which could severely harm our pi and sensors.



One thing we didn't take into consideration before, was the bms being in the battery pack, upon opening it we discovered that the bms had melted through the casing with light usage, it needed to be moved away, unfortunately because we had such an old 3d printer we couldn't print anything besides pla which softens at 60 degrees and will start deforming and smelling not so nice, so we designed a quick blower fan cooler rig running only on 5 watts keeping the bms cool as ice.

In terms of calibration, the only part that requires calibration is the steering servo because it doesn't automatically know its current angle on start it has to be manually calibrated using a ruler and a simple calibration program, although the code will also make of with any miscalibrations with corrections using the walls of the arena.

Placement of the pi was a bit tough, putting it on its main surface would be very space consuming so we decided a stacked vertical orientation would be perfect, saving space, and allowing for far better airflow, although making connectivity a lot harder.